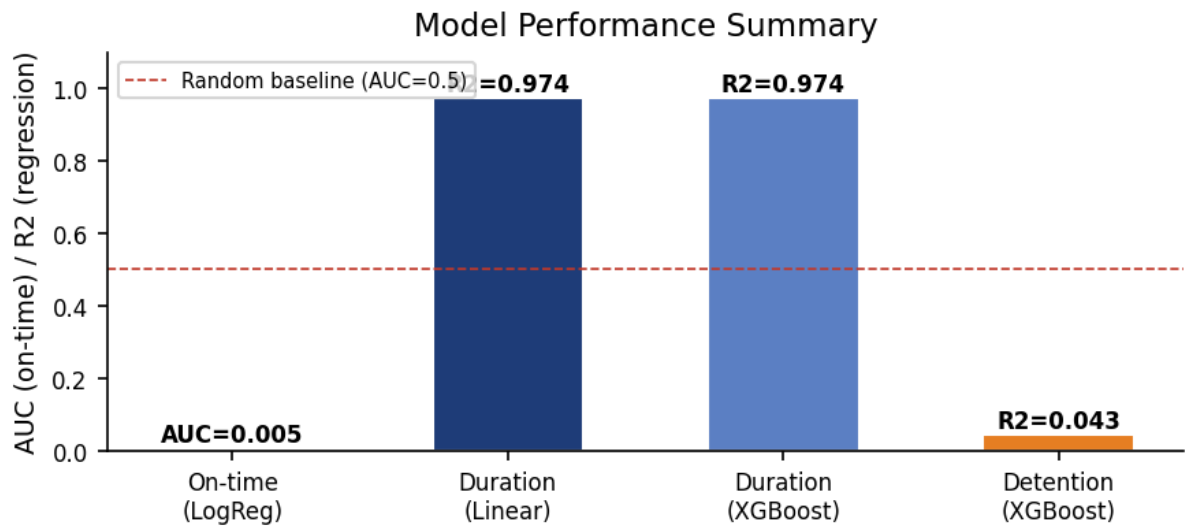


Logistics Operations Database

Analysis & Predictive Modeling Report

Dataset: yogape/logistics-operations-database | 549,706 records | 2022-2024
June 2026

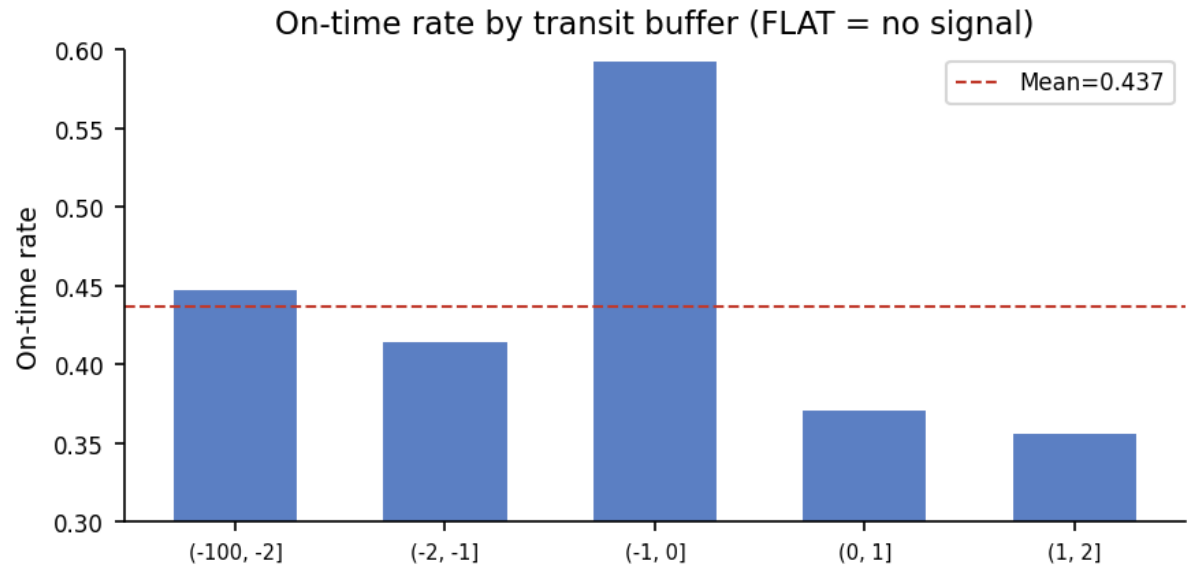
Three predictive targets were modeled: on-time delivery (AUC 0.505 -- random noise), trip duration (R2 0.974 -- distance-driven), and detention minutes (R2 0.04 -- weak signal from event type and time of day). Driver profile carries no predictive signal (ICC ~0.001). A fuel_purchases data bug was found and corrected, revealing 72% fleet margins.



1. Dataset & On-Time Delivery

14 tables, 549,706 records (2022-2024). Target: on_time_flag (delivery events, 85,410 rows). Logistic regression with chronological split (train 70,973 / test 14,437).

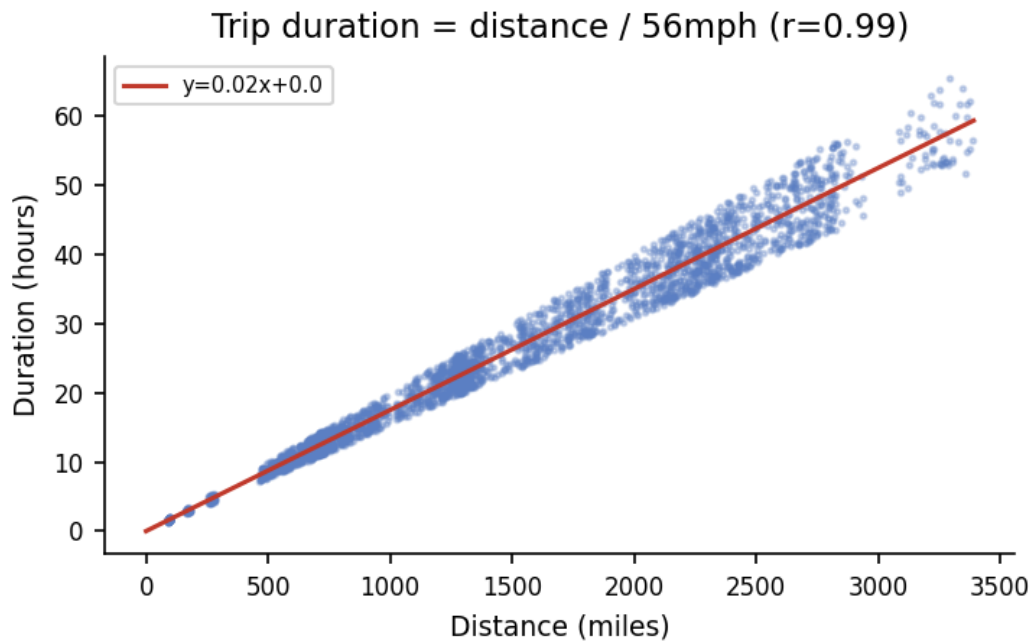
Metric	Value	Interpretation
AUC	0.505	= coin flip (no signal)
Accuracy	0.508	Below 0.557 majority baseline
Top feature corr	0.007	Effectively zero



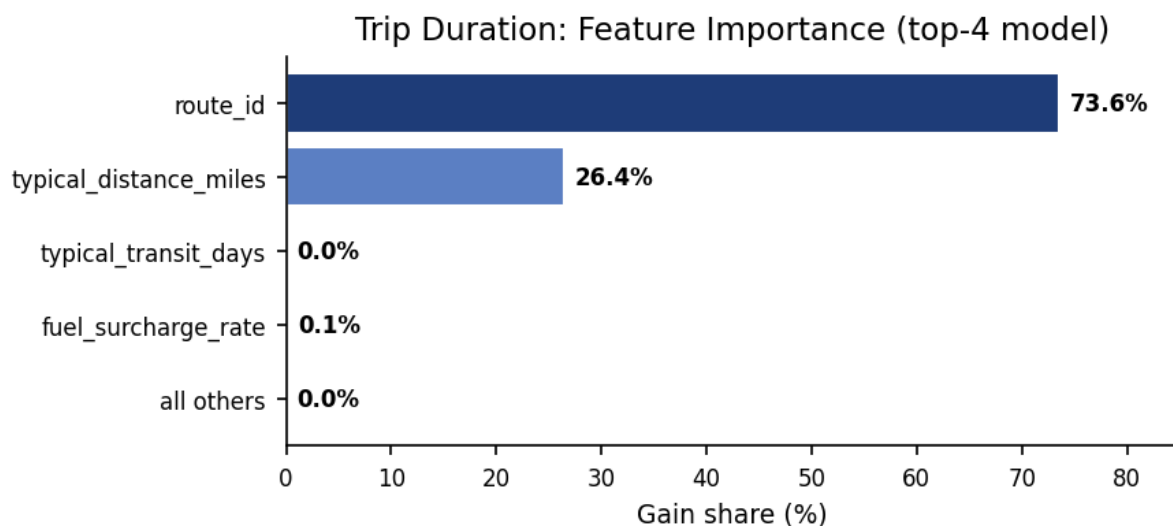
KEY FINDING: on_time_flag is random Bernoulli(0.45) noise. Flat across every cut: transit buffer, route, customer, month, lead time. Even a timestamp-derived label is flat. Unpredictable.

2. Trip Duration Prediction

Target: actual_duration_hours (85,410 trips). Linear & XGBoost, chronological split. typical_distance_miles replaces actual_distance_miles (leak-safe, $r=0.998$).



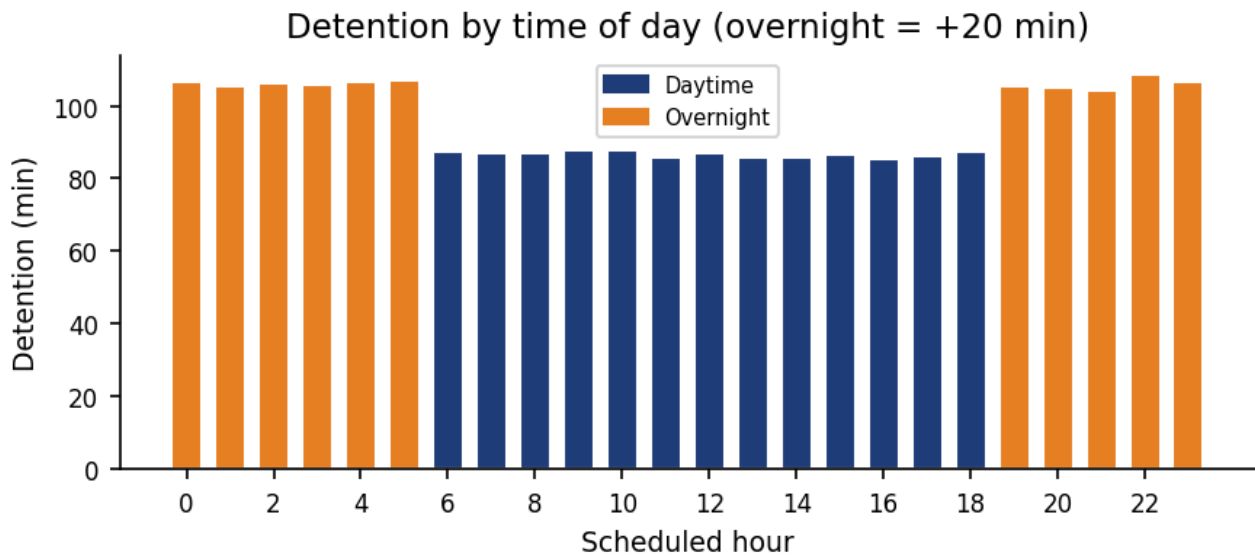
Model	R2	MAE (h)	RMSE (h)
Linear (59 feats)	0.974	1.698	2.300
XGBoost (59 feats)	0.973	1.714	2.336
XGBoost (top 4)	0.974	1.694	2.299
distance/mph baseline	0.974	1.689	2.305



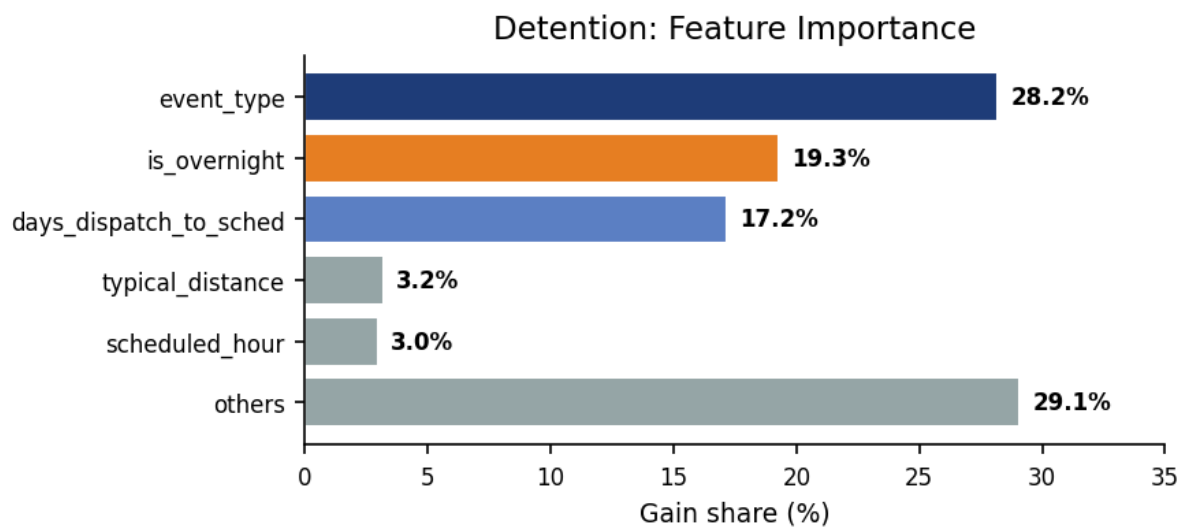
KEY FINDING: Duration = distance / 56mph + noise. route_id + typical_distance_miles = 100% of gain. Dropping 55 features improved the model. No nonlinear structure to exploit.

3. Detention Minutes Prediction

Target: detention_minutes (170,820 events, both pickup+delivery). Not distance-driven ($r=0.001$). XGBoost, chronological split (train 127,617 / test 28,780).



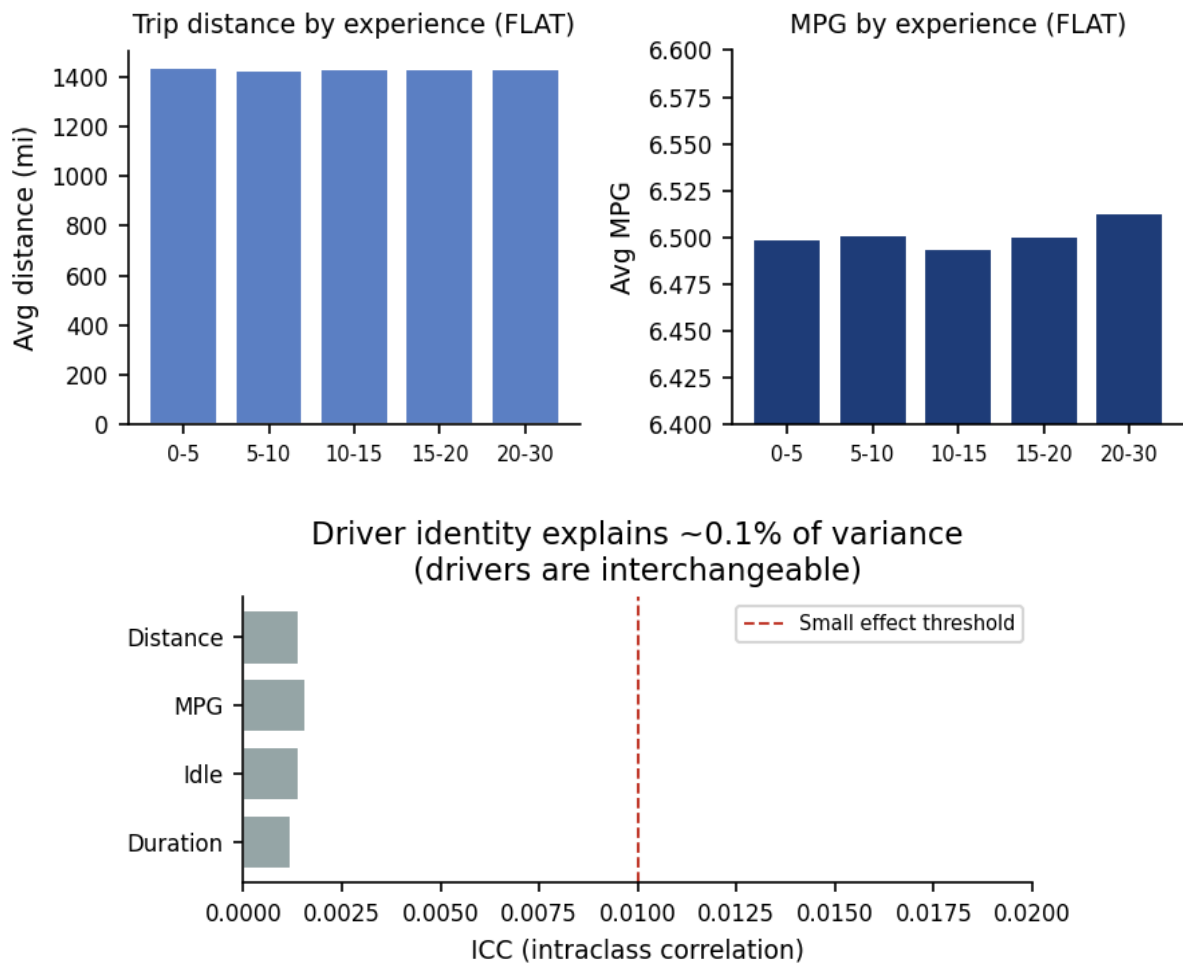
Model	R2	MAE (min)
XGBoost (75 feats)	0.043	58.1
event_type mean	0.047	58.0
predict train mean	-0.000	59.0



KEY FINDING: Real but weak signal ($R^2=0.04$). Deliveries +30 min vs pickups; overnight +20 min; short lead time increases detention. 96% of variance unexplained -- needs features not in dataset (dock windows, staffing, weather).

4. Driver Profile: No Signal

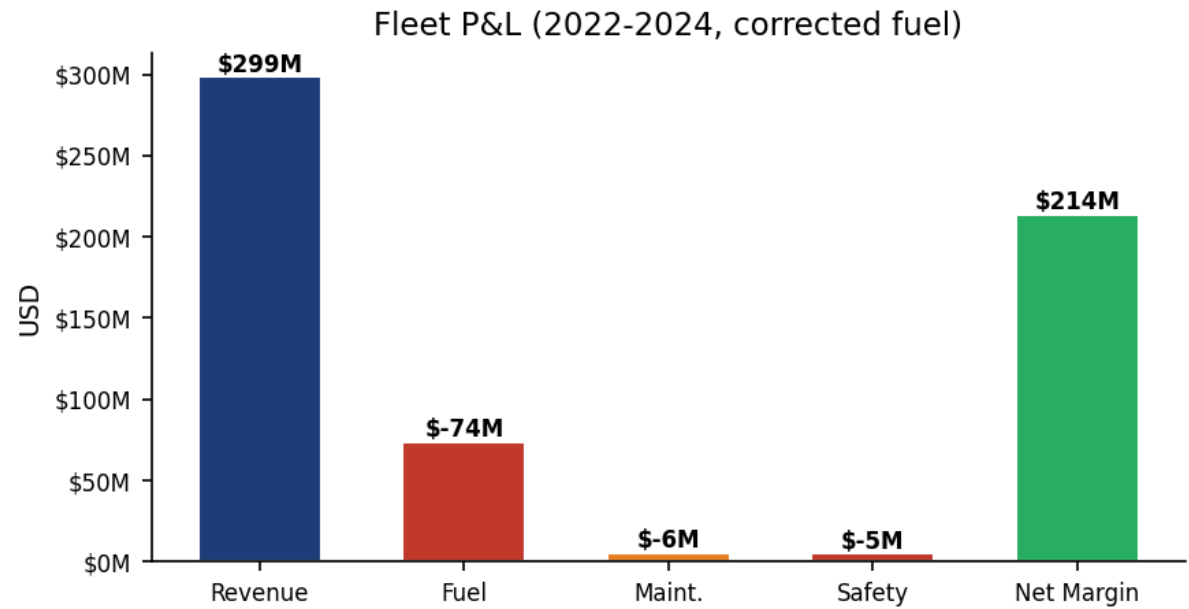
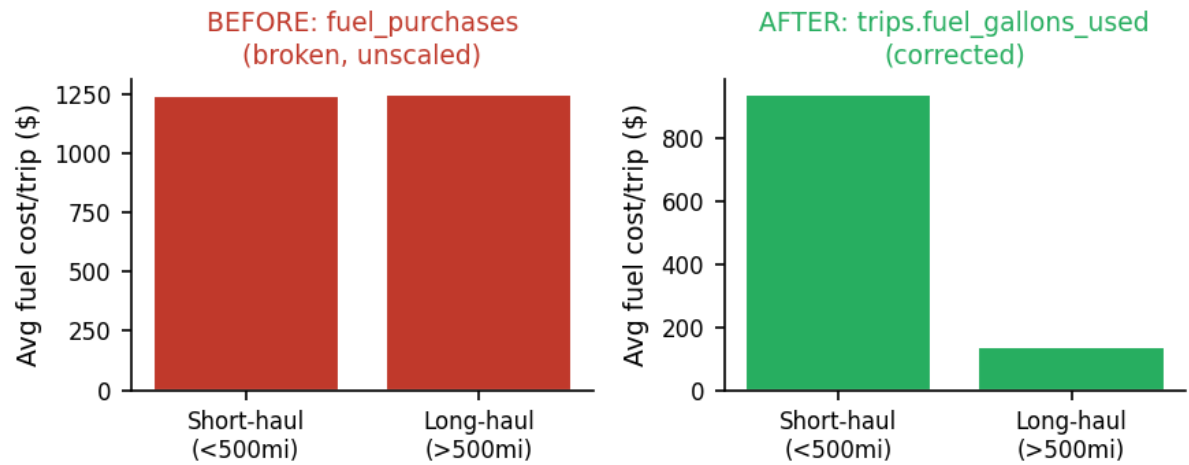
Tested: years_experience, age, tenure, CDL, home_terminal, license_state vs all outcomes. 4 grouping strategies, ANOVA, ICC, trip-assignment randomness.



KEY FINDING: Driver identity explains 0.1% of variance (ICC ~0.001). Trip assignment is random w.r.t. experience (same distances, routes, trucks for all). Profile is decorative.

5. Data Bug & Profitability

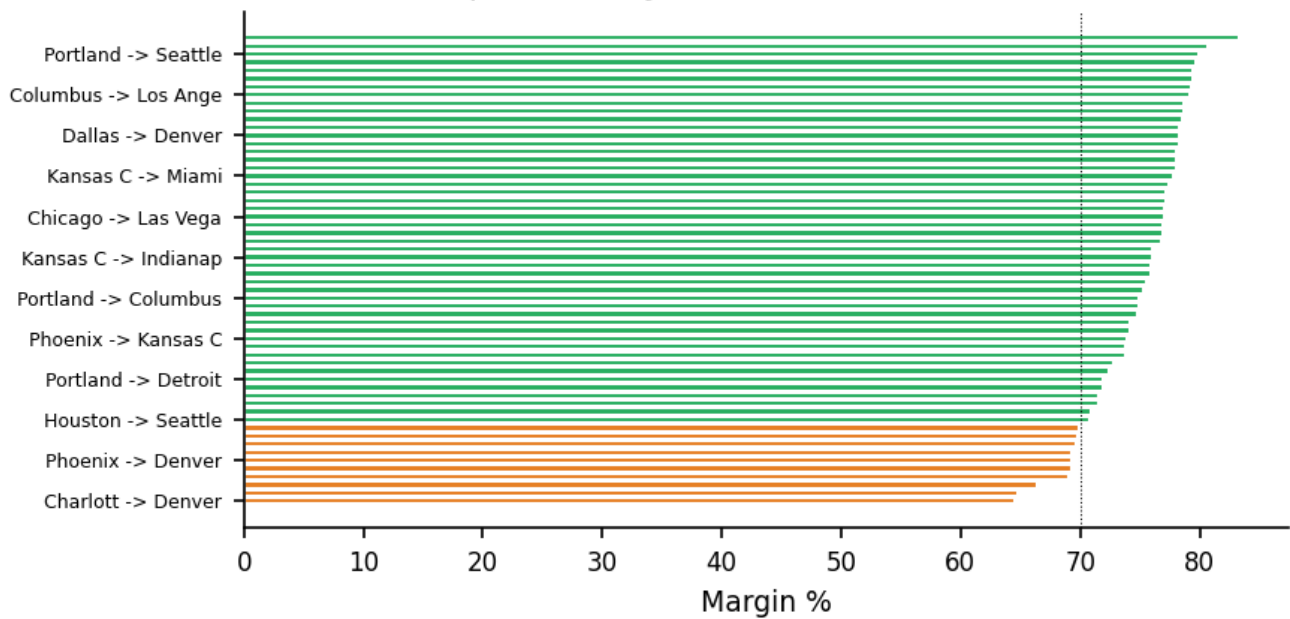
fuel_purchases.csv gallons are unscaled to trip distance (data-gen bug). 92-mile trips get same fuel bill as 2,500-mile trips. Fix: use trips.fuel_gallons_used x \$3.90/gal.



KEY FINDING: Bug inflated fuel by \$22M. After correction: all 58 routes profitable, fleet margin 72% (\$215M), fuel 24.7% of revenue (was 32%).

6. Routes & Seasonality

Route profitability (all 58 routes, corrected fuel)



All 58 routes profitable (corrected). Short-haul (92-167 mi) lowest margins (50-78%); long-haul (2,500+ mi) highest (75-80%).

Monthly load volume (no seasonality)



No seasonality: ~2,300-2,500 loads/month, ~\$3,070 avg revenue, ~\$2.20/mile rate, every month. Margins stable at ~72% throughout 2022-2024.

7. Conclusions

Target	Best Model	Performance	Signal
On-time delivery	LogReg	AUC 0.505	None (noise)
Trip duration	XGBoost top-4	R2 0.974	Distance (100%)
Detention minutes	XGBoost	R2 0.043	Event type, time (4%)

This is a synthetic dataset where distance is the only real driver of trip outcomes, on-time is random noise, and driver profile is disconnected from performance. A fuel data bug masked true profitability (72% margins, all routes profitable). For real-world logistics prediction, the valuable targets would be detention (with dock/staffing/weather features) and fuel efficiency (with driver-skill variation).